

AI-Powered Web-Based Data Analysis and Prediction Platform

A research analytics application for econometric modeling and forecasting

Document type	Official Application User Guide
Audience	Researchers, economists, and data analysts
Access	Web browser (no installation required)
Sign-in	Google Account
Version	1.0

Table of Contents

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1. Application Overview

The **AI-Powered Web-Based Data Analysis and Prediction** platform is an online research tool that lets you run professional statistical and machine-learning models on your own datasets — **without writing any code**.

You upload a spreadsheet (a CSV file), point and click to tell the application which columns are your outcome and your predictors, choose a statistical method, and the platform estimates the model on secure servers and returns your results in clear tables and charts. Results can be downloaded as a ready-to-present PowerPoint deck.

The application is organized around the three data structures most common in economic and social-science research:

Module	Use it when your data is...
Cross-Sectional Data	Many subjects observed once (e.g. 500 firms surveyed this year)
Time Series Data	One subject observed repeatedly over time (e.g. monthly inflation for 10 years)
Panel Data	Many subjects observed over multiple time periods (e.g. 50 countries over 20 years)

A fourth module, **Data Cleanup**, helps you prepare a messy file before analysis.

Note: This is a research and estimation tool. It supports your analysis and reporting; it does not replace your judgment as a researcher about which model is appropriate for your question.

2. Purpose and Scope

Purpose

The platform exists to make rigorous econometric estimation **accessible, fast, and reproducible** for researchers who understand statistics but may not program. It removes the software-engineering barrier — no Python, R, or Stata installation, no package management, no scripting — while still providing the diagnostic depth that credible empirical work requires.

In scope

- Uploading tabular research data (CSV format).
- Descriptive/summary statistics for selected variables.
- Estimating regression and machine-learning models across cross-sectional, time-series, and panel data.
- Automatic model diagnostics (tests for heteroskedasticity, multicollinearity, autocorrelation, normality, model specification).
- Forecasting and prediction with out-of-sample validation.
- Visualizing a time series as a line graph.
- Basic data cleaning (handling missing values, converting and normalizing columns).

- Exporting formatted results to PowerPoint.

Out of scope

- The platform does **not** collect data for you or clean it beyond the operations in the Data Cleanup module.
- It does **not** choose the "correct" model for your research question — you select the method.
- It is **not** a general-purpose spreadsheet, database, or data-warehousing tool.
- It does **not** currently accept file formats other than CSV.

3. Intended Users and Use Cases

Who this guide is for

This guide assumes you are a **researcher, economist, graduate student, or analyst** who:

- Understands core statistical ideas (regression, significance, R^2 , etc.).
- Works with structured, tabular data.
- Wants credible results quickly, without programming.

No programming experience is assumed or required.

Representative use cases

User	Typical goal
Applied economist	Estimate the effect of education on wages using survey data (cross-sectional OLS).
Macro/finance researcher	Forecast a price or index over the next periods (time series ARIMA).
Development/policy researcher	Measure a policy's impact across countries over time (panel fixed effects).
Graduate student	Produce diagnostic-checked regression tables and slides for a thesis chapter.
Data analyst	Compare several models (OLS vs. LASSO vs. Random Forest) on the same dataset.

4. Key Concepts and Terminology

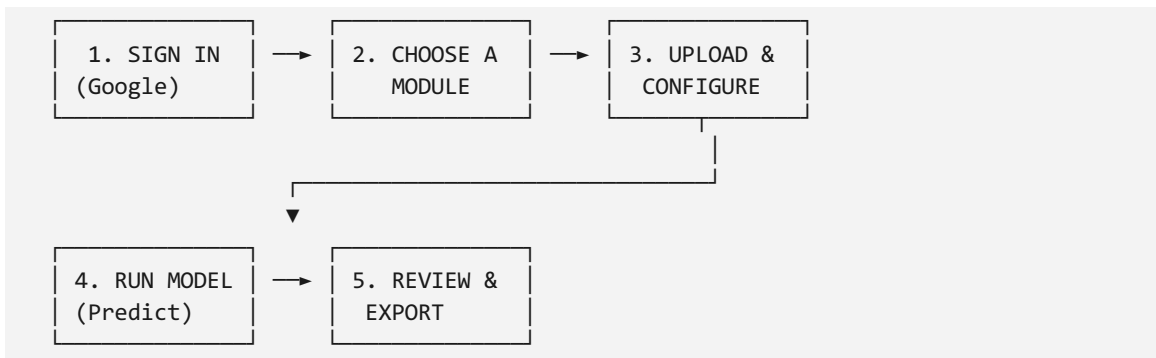
To keep the rest of this guide clear, here are the terms used consistently throughout, in plain language.

Term	Plain-language meaning
Dependent variable	The outcome you are trying to explain or predict (e.g. wage, price, GDP). Also called the <i>*endogenous*</i> or <i>*target*</i> variable.
Independent variable	A predictor you believe influences the outcome (e.g. years of schooling). Also called a <i>*feature*</i> or <i>*regressor*</i> .

Exogenous variable	In time series, an outside predictor used alongside time (e.g. temperature when forecasting energy demand).
ID column	The column that uniquely labels each subject (person, firm, country). Used to identify rows, not as a predictor.
Categorical variable	A predictor made of categories rather than numbers (e.g. region, industry). The app converts these to numbers automatically.
Coefficient	A number showing how much the outcome changes when a predictor increases by one unit.
p-value	The probability the observed relationship is due to chance. Small (below 0.05) usually means "statistically significant."
R² (R-squared)	The share of variation in the outcome explained by the model, from 0 to 1. Higher generally means a better fit.
Feature importance	For tree-based models, a ranking of which predictors mattered most (used in place of coefficients).
Diagnostic test	An automatic check on whether the model's statistical assumptions hold.
Cross-validation (CV)	Repeatedly testing the model on data it did not train on, to gauge how well it generalizes.

5. System Workflow Overview

Every analysis in the platform follows the same five-stage flow, regardless of module:



1. **Sign in** with your Google Account.
2. **Choose a module** from the dashboard that matches your data structure.
3. **Upload** your CSV and **configure** the analysis (select method and variables).
4. **Run** summary statistics, a line graph, and/or the model (**Predict**).
5. **Review** the tables and diagnostics, then **export** to PowerPoint.

Tip: Every analysis screen has a "**Guide Me**" button in the top-right corner that opens step-by-step help specific to that screen. This user guide expands on that in-app help.

6. Getting Started: Signing In

The platform requires a Google sign-in. This protects your session and your data.

Procedure — Signing in

6. Open the application in your web browser.
7. On the **Welcome** screen, locate the **"Sign in with your Google Account"** button.
8. Click the Google sign-in button.
9. In the Google window, choose the account you want to use and approve access.
10. On success, you are taken automatically to the **Dashboard**.

What you should expect: Your first name and email appear in a small user card in the top-right of the dashboard, confirming you are signed in.

Note: If your session expires while you work, the application may ask you to sign in again. This is normal security behavior — simply sign back in and continue.

Signing out

Click the red **Logout** button in the user card (top-right). You are returned to the Welcome screen.

7. The Dashboard

After signing in you arrive at the **Dashboard**, titled **"AI-Powered Web-Based Data Analysis and Prediction."**

What you see

- A heading and the prompt **"Select a data category to continue."**
- Three large clickable cards:
 - **Cross-Sectional Data**
 - **Time Series Data**
 - **Panel Data**
- A **"Guide Me"** button (top-right) with dashboard-level help.
- Your **user card** with your name, email, and **Logout**.

Procedure — Opening a module

11. Decide which data structure matches your dataset (see the table in [Section 1](#1-application-overview)).
12. Click the matching card.
13. The module's analysis screen opens, ready for you to upload a file.

To return to the dashboard from any module, click the **back arrow** (←) in the top-left of the module's header.

8. Preparing Your Data

Good input data is the single biggest factor in getting trustworthy results. Please read this section before your first upload.

File format requirements

Requirement	Detail
File type	CSV (comma-separated values). Export from Excel/Sheets using *Save As → CSV*.
First row	Must contain column names (headers) . These names populate the app's dropdown menus.
One observation per row	Each row is a single record (a person, firm, country, or a single time point).
Consistent columns	Every row should have the same set of columns.

Content guidelines

- **Use clear, unique column names** (e.g. wage, education_years, region) — you will select them by name in dropdowns.
- **Numbers should be numeric**, not text. Avoid currency symbols, thousands separators, or stray spaces inside numeric columns (e.g. use 1250, not \$1,250).
- **Dates** (for time series and panel data) should be in a recognizable format such as 2023-01-31, 01/31/2023, or Jan 2023. The app validates that the column can be read as dates.
- **Missing values** are allowed, but each module handles them; consider using the **Data Cleanup** module first (see [Section 12](#12-module-data-cleanup)).

Data-size guidance

- Time-series estimation requires enough history to validate the model. As a rule of thumb, provide at least **10 observations**, and more when you increase the number of cross-validation folds. If a dataset is too short, the app will tell you the minimum it needs.
- Very large files take longer to upload and estimate. Start with a focused dataset containing only the columns you need.

Tip: Before running a model, always click "**Summary Statistics**" first. It's a fast sanity check that your columns were read correctly (reasonable means, min/max, and counts).

9. Module: Cross-Sectional Data Analysis

Use this module when you have **many subjects observed at a single point in time** — for example, a survey of 1,000 households this year.

9.1 Available methods

Method (menu label)	Best used for
OLS	Standard linear regression for a continuous outcome. The default starting point.
General Least Square (GLS)	Linear regression when errors are unequal or correlated.

Logit	A binary outcome (yes/no, 1/0), e.g. employed vs. not.
LASSO	Regression with many predictors; automatically shrinks weak ones toward zero.
RIDGE	Regression with many correlated predictors; stabilizes the estimates.
FOREST	Random Forest — a flexible tree-based model for non-linear relationships.
BOOSTING	Gradient Boosting — a powerful tree-based model built in stages.
BAGGING	An ensemble of trees averaged to reduce variance.

9.2 Screen elements

Element	What it does
Upload File	Opens a file picker; select your .csv. Column names load into the dropdowns.
Methods	Choose the estimation method (table above).
ID Column	The unique row identifier (not used as a predictor).
Dependent Variable	The outcome you want to explain or predict.
Independent Variables	One or more predictors (multi-select).
Categorical Variables	Any predictors that are categories; encoded automatically (multi-select).
Outliers	Choose Yes to detect and treat extreme values before estimating, or No to leave data as-is.
Summary Statistics (button)	Shows descriptive stats for your selected variables.
Predict (button)	Runs the model. Becomes active only when required fields are set.
Download as PowerPoint	Appears after a successful run; exports a formatted deck.

9.3 Procedure — Running a cross-sectional model

- Click a **Cross-Sectional Data** card on the dashboard.
- Click **Upload File** and select your CSV. Wait for the dropdowns to populate.
- In **Methods**, choose your estimation method (e.g. **OLS**).
- Set the **ID Column** to your unique identifier.
- Set the **Dependent Variable** to your outcome.
- Add one or more **Independent Variables**.
- If any predictors are categories (e.g. region), add them under **Categorical Variables**.
- Choose an **Outliers** option (**Yes** or **No**).
- *(Recommended)* Click **Summary Statistics** and confirm the numbers look sensible.
- Click **Predict**. A loading indicator appears while the model runs.
- Review the results table and diagnostics ([Section 13](#13-understanding-your-results)).
- *(Optional)* Click **Download as PowerPoint**.

Note: The **Predict** button stays greyed-out until an ID column, dependent variable, at least one independent variable, and an outlier choice are all selected. This prevents incomplete runs.

10. Module: Time Series Analysis

Use this module when you have **repeated observations over time** — for example, monthly sales or daily prices.

10.1 Available methods

Method	Best used for
ARIMA	The classic forecasting model for trend and seasonality over time.
LASSO / RIDGE	Regularized regression using time and optional outside predictors.
FOREST / BOOSTING / BAGGING	Tree-based ensemble models adapted for time-ordered data.

10.2 Screen elements

Element	What it does
Upload File	Select your .csv; column names load into the dropdowns.
Methods	Choose the forecasting method (table above).
Date Variable	The column holding dates/timestamps. The app checks it contains valid dates.
Start Date / End Date	Populate automatically from your data; narrow the range to model a specific period.
Endogenous Variable	The target you want to forecast.
Exogenous Variables	Optional additional predictors (multi-select).
Summary Statistics (button)	Descriptive stats for the selected variables.
Line Graph (button)	Plots your target variable over time.
Predict (button)	Runs the forecast; active only when required fields are set.
Clear (button)	Resets the whole screen for a new dataset.
Download	Exports the forecast to PowerPoint after a run.

10.3 Procedure — Producing a forecast

26. Open the **Time Series Data** module.

27. **Upload** your CSV.

28. Choose a **Method** (e.g. **ARIMA**).

29. Select the **Date Variable**.

- If the column isn't valid dates, a red warning appears and you cannot proceed. Pick a different column or fix the data.

30. Confirm or adjust the **Start Date** and **End Date** (they default to your full range).

31. Select the **Endogenous Variable** (your forecast target).

32. *(Optional)* Add **Exogenous Variables**.
33. *(Recommended)* Click **Line Graph** to visually inspect the series for trends, gaps, or spikes, and **Summary Statistics** for a numeric check.
34. Click **Predict**.
35. Review the forecast and diagnostics.
36. *(Optional)* **Download** the PowerPoint.
37. Click **Clear** to start over with a new file.

Tip: The line graph is the fastest way to catch a data problem — a flat line, a sudden jump, or a gap usually signals a formatting or missing-value issue worth fixing before forecasting.

11. Module: Panel Data Analysis

Use this module when you have **many subjects observed across multiple time periods** — for example, 40 firms tracked annually for 15 years.

11.1 Available methods

Method	Best used for
Fixed Effects	Controls for stable, unobserved traits of each subject that don't change over time.
Random Effects	Assumes subject-level differences are uncorrelated with your predictors.
LASSO / RIDGE	Regularized regression with many predictors.
FOREST / BOOSTING / BAGGING	Tree-based ensemble models.

11.2 Screen elements

Panel analysis combines cross-sectional and time controls:

Element	What it does
Upload File	Select your .csv.
Methods	Choose the estimation method.
Id of the Data	The column identifying each subject (firm/person/region). Required to separate subjects.
Dependent Variable	The outcome.
Independent Variables	Predictors (multi-select).
Categorical Variables	Category predictors, encoded automatically.
Outliers	Detect/treat extreme values, or leave as-is.
Date Variable	The time-period column (year, quarter, month). This is what makes the data a panel.
Summary Statistics / Predict / Clear	As in the other modules.
Download	Exports results to PowerPoint.

11.3 Procedure — Running a panel model

38. Open the **Panel Data** module.
39. **Upload** your CSV.

40. Choose a **Method** (e.g. **Fixed Effects**).
41. Set **Id of the Data** to your subject identifier.
42. Set the **Dependent Variable**.
43. Add **Independent Variables** (and **Categorical Variables** if any).
44. Choose an **Outliers** option.
45. Select the **Date Variable** (the time column). If it isn't valid dates, a warning appears.
46. *(Recommended)* Click **Summary Statistics**.
47. Click **Predict**.
48. Review results and diagnostics; **Download** if desired.
49. Use **Clear** to reset.

Note: The key difference from cross-sectional analysis is the **Date Variable**: the **same** subject ID appears on multiple dates. Make sure your ID and date columns are both correct, or the panel structure won't be recognized.

12. Module: Data Cleanup

Before modeling, use the **Data Cleanup** module to fix common data problems. It works on the file in your browser and shows you an immediate preview.

12.1 Cleanup actions

Action (menu label)	What it does
Drop Missing Values	Removes the selected column.
Convert Text to Numeric	Turns number-like text into true numbers.
Encode Text as Categorical Dummies	Rescales the column's numeric values to a 0–1 range (normalization).
Fill Missing with Global Average	Replaces blank cells with the column's overall average.
Fill Missing with Interpolated Average	Fills a blank using the values immediately before and after it.

12.2 Screen elements

Element	What it does
Upload File	Select your .csv.
Variables	Choose the single column to clean.
Type of Data Cleanup	Choose the action to apply (table above).
Summary Statistics (button)	Shows descriptive stats for all columns.
Apply Cleanup (button)	Applies the action; active only when a column and action are chosen.
Clear (button)	Resets the screen.
Preview table	After cleanup, shows the first 20 rows so you can verify the change.

12.3 Procedure — Cleaning a column

50. Open the **Data Cleanup** module.

51. **Upload** your CSV.
52. In **Variables**, select the column to fix.
53. In **Type of Data Cleanup**, choose the action.
54. Click **Apply Cleanup**.
55. Inspect the **preview table** and the updated summary statistics.
56. Repeat for other columns as needed.

Important: Cleanup here operates on the copy loaded in your browser for inspection. Treat it as a preparation and preview step: verify the effect in the preview, then use a cleaned source file in the analysis modules for your final run. Always keep an untouched copy of your original data.

13. Understanding Your Results

After you click **Predict**, the platform returns a results view. What appears depends on the method you chose.

13.1 For regression methods (OLS, GLS, Logit, LASSO, Ridge, panel models)

You will typically see:

Output	How to read it
Coefficient	The estimated effect of each predictor on the outcome. Sign matters: positive = raises the outcome, negative = lowers it.
Standard error	The uncertainty around each coefficient. Smaller is more precise.
p-value	Below 0.05 is commonly treated as statistically significant .
Significance label	The app labels each result " significant " or " not significant " at the 0.05 threshold.
Fit statistics (e.g. R^2)	How much of the outcome's variation the model explains.

13.2 For tree-based methods (Forest, Boosting, Bagging)

Tree models don't produce coefficients. Instead you'll see:

- **Feature importance** — a ranking of which predictors contributed most to the predictions.
- **Performance metrics** — such as R^2 , **RMSE** (root mean squared error), and **MAE** (mean absolute error). For RMSE and MAE, **smaller is better**.

13.3 Diagnostic tests (regression models)

The platform runs standard assumption checks automatically and states a plain-language conclusion for each:

Test	Question it answers	What a "flag" means
Breusch-Pagan / White	Is the error spread constant (homoskedastic)?	If detected, the guide suggests robust standard errors.

Variance Inflation Factor (VIF)	Are predictors too correlated (multicollinearity)?	High VIF means predictors overlap; consider dropping one.
Durbin–Watson	Are residuals autocorrelated?	Values far from ~2 suggest correlation in the errors.
Jarque–Bera	Are residuals normally distributed?	A significant result means non-normal residuals.
RESET	Is the model's functional form correct?	A significant result suggests a missed non-linearity.

13.4 Forecast and cross-validation output (time series)

- **Forecasted values** for the target variable.
- **Cross-validation metrics** reported as an average (mean) and a spread (standard deviation) across folds — this tells you how *stable* the model's out-of-sample accuracy is. A small spread means dependable performance.

Tip: Report diagnostics alongside your coefficients. Reviewers of empirical work expect to see that assumptions were checked, not just the headline results.

14. Exporting Results to PowerPoint

Every analysis module can export results to a formatted PowerPoint deck, ready to drop into a paper, thesis, or presentation.

Procedure — Exporting

57. Run a successful **Predict** (or **Summary Statistics**).
58. Locate the **download** button that appears above/near the results table (labeled **Download as PowerPoint** or a download icon).
59. Click it. Your browser downloads a .pptx file.
60. Open it in PowerPoint, Google Slides, or Keynote.

What the deck contains: your model summary, coefficients or feature importance, fit statistics, and diagnostic tests (for model runs), or the descriptive table (for summary statistics).

Tip: Run **Summary Statistics** and **Predict** separately if you want both a descriptive-statistics slide and a results slide to export.

15. Worked Examples

The following realistic scenarios illustrate end-to-end use. Substitute your own data and columns.

Example A — Returns to education (Cross-Sectional, OLS)

Research question: *Does an additional year of schooling raise hourly wages?*

Data: One row per worker, with columns `worker_id`, `wage`, `education_years`, `experience`, `region`.

Steps:

61. Open **Cross-Sectional Data** → **Upload** the CSV.
62. **Method:** OLS. **ID Column:** worker_id. **Dependent Variable:** wage.
63. **Independent Variables:** education_years, experience. **Categorical Variables:** region.
64. **Outliers:** Yes. Click **Summary Statistics** to confirm wages look sensible.
65. Click **Predict**.

Expected result: A positive, statistically significant coefficient on education_years would indicate that more schooling is associated with higher wages, holding experience and region constant. Check the Breusch–Pagan test; if heteroskedasticity is flagged, note that robust standard errors are advisable.

Example B — Forecasting monthly rental prices (Time Series, ARIMA)

Research question: *What is the expected trajectory of average rents?*

Data: One row per month, with month (dates) and avg_rent.

Steps:

66. Open **Time Series Data** → **Upload**.
67. **Method:** ARIMA. **Date Variable:** month. Confirm the green validation (no warning).
68. Leave **Start/End Date** at the full range. **Endogenous Variable:** avg_rent.
69. Click **Line Graph** to eyeball the trend, then **Predict**.

Expected result: Forecasted rent values plus cross-validation metrics. A low standard deviation across folds means the forecast is stable; a high one means treat the forecast with caution.

Example C — Effect of a policy across countries (Panel, Fixed Effects)

Research question: *Did a tax reform change investment, controlling for fixed country traits?*

Data: One row per country-year, with country_id, year, investment, tax_rate, gdp_growth.

Steps:

70. Open **Panel Data** → **Upload**.
71. **Method:** Fixed Effects. **Id of the Data:** country_id. **Date Variable:** year.
72. **Dependent Variable:** investment. **Independent Variables:** tax_rate, gdp_growth.
73. **Outliers:** No. **Summary Statistics**, then **Predict**.

Expected result: The coefficient on tax_rate estimates its effect on investment *within* countries over time, netting out stable country differences.

16. Common Errors and Troubleshooting

Symptom	Likely cause	What to do
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Predict button is greyed out	A required field is missing.	Ensure the method, ID/date column, dependent variable, and at least one independent variable (and outlier choice, where shown) are all selected.
"Selected column does not contain valid date values"	The chosen Date Variable has non-date entries.	Pick the correct date column, or reformat the dates (e.g. 2023-01-31) and re-upload.
"Something went wrong. Try again!!"	The model could not be estimated with the current inputs.	Check for too many missing values, non-numeric data in numeric columns, or too few rows. Clean the data and retry.
"Not enough observations..."	The dataset is too short for the requested validation.	Provide more rows, or reduce the number of cross-validation folds if available.
"Too many requests. Please try again in one minute."	You ran many analyses in quick succession (rate limit).	Wait about a minute, then continue. Space out large batches of runs.
Asked to sign in again / "Missing or invalid authorization"	Your session expired.	Sign in again with Google and re-run.
Dropdowns are empty after upload	The file isn't a valid CSV or lacks a header row.	Re-export as CSV with column names in the first row.
Numbers look wrong in Summary Statistics	Numeric columns contain symbols or text.	Use Data Cleanup → *Convert Text to Numeric*, or clean the source file.
Empty or flat Line Graph	Wrong date/target column, or missing values.	Re-check the Date Variable and Endogenous Variable selections.

Note: If an error persists after checking the table above, click **Clear**, re-upload a small, clean subset of your data, and confirm it runs. Then add complexity back gradually to isolate the problem.

17. Best Practices for Researchers

Follow these recommendations to get results you can defend.

Before you model

74. **Keep a pristine original.** Never overwrite your raw data file; clean on copies.
75. **Name columns clearly.** You'll select them by name — clarity prevents mistakes.
76. **Inspect first.** Always run **Summary Statistics** (and **Line Graph** for time series) before **Predict**.
77. **Fix, don't ignore, missing data.** Use the Data Cleanup module deliberately and document what you did.

Choosing a method

78. **Match the model to the data structure.** Cross-sectional, time series, and panel are not interchangeable — use the right module.

79. **Match the model to the outcome.** Use **Logit** for yes/no outcomes; **OLS/GLS** for continuous outcomes.
80. **Compare models.** Run OLS, then LASSO/Ridge, then a tree-based model on the same data and compare fit and stability before concluding.

Interpreting results

81. **Read the diagnostics, not just the coefficients.** If heteroskedasticity or multicollinearity is flagged, adjust your interpretation accordingly.
82. **Distinguish significance from importance.** A significant coefficient can still be small; look at the magnitude, not only the p-value.
83. **Prefer stable forecasts.** In time series, trust models whose cross-validation spread is small.

Reporting

84. **Export as you go.** Download the PowerPoint immediately after a good run so your results and diagnostics stay together.
85. **Record your settings.** Note the method, variables, date range, and outlier choice for each result so it's reproducible.

18. Quick Reference

Module selector

Your data	Module	Common first method
Many subjects, one time point	Cross-Sectional Data	OLS
One subject, many time points	Time Series Data	ARIMA
Many subjects, many time points	Panel Data	Fixed Effects
Needs cleaning first	Data Cleanup	—

Buttons at a glance

Button	Effect
Summary Statistics	Descriptive stats for selected variables.
Line Graph *(time series)*	Plots the target over time.
Predict	Estimates the model and shows results.
Apply Cleanup *(cleanup)*	Applies the chosen cleaning action.
Clear	Resets the screen for a new dataset.
Download as PowerPoint	Exports the current results to .pptx.
Guide Me	Opens on-screen help for the current screen.
Logout	Ends your session.

Result cheat-sheet

You see...	It means...
p-value < 0.05	Statistically significant (at the 5% level).
High R²	Model explains much of the outcome's variation.
Low RMSE / MAE	Predictions are close to actual values.

Feature importance	Which predictors mattered most (tree models).
Diagnostic "detected"	An assumption may be violated — interpret with care.

End of User Guide.